IEC 61754-7-2

Edition 1.0 2017-12

INTERNATIONAL STANDARD

Fibre optic interconnecting devices and passive components – Fibre optic connector interfaces –
Part 7-2: Type MPO connector family – Two fibre rows

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9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

International Standard IEC 61754-7-2 has been prepared by subcommittee 86B: Fibre optic interconnecting devices and passive components, of IEC technical committee 86: Fibre optics.

This first edition of IEC 61754-7-2, along with the first edition of IEC 61754-7-1, cancels and replaces the third edition of IEC 61754-7 published in 2008.

This first edition of IEC 61754-7-2 includes the two fibre row MPO variants including the addition of active device receptacles and up-angled plugs.

The first edition of IEC 61754-7-1 includes the one fibre row MPO variants and related active device receptacles and up-angled plugs.

Following the publication of both IEC 61754-7-1 and IEC 61754-7-2, IEC 61754-7 will be withdrawn.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
86B/4099/FDIS	86B/4110/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all the parts in the IEC 61754 series, published under the general title *Fibre optic interconnecting devices and passive components – Fibre optic connector interfaces*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

The International Electrotechnical Commission (IEC) draws attention to the fact that it is claimed that compliance with this document may involve the use of patents concerning MPO connectors.

The IEC takes no position concerning the evidence, validity and scope of these patent rights.

The holders of these patent rights have assured the IEC that they are willing to negotiate licences under reasonable and non-discriminatory terms and conditions with applicants throughout the world. In this respect, the statement of the holders of these patent rights is registered with the IEC. Information may be obtained from:

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FIBRE OPTIC INTERCONNECTING DEVICES AND PASSIVE COMPONENTS – FIBRE OPTIC CONNECTOR INTERFACES –

Part 7-2: Type MPO connector family – Two fibre rows

1 Scope

This part of IEC 61754 defines the standard interface dimensions for the type MPO family of connectors with two rows of fibres.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

There are no normative references in this document.

3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

4 Description

The parent connector for the type MPO connector family is a multiway plug connector characterized by a rectangular ferrule normally 6,4 mm × 2,5 mm which utilizes two pins of 0,7 mm diameter as its alignment. The variant in this document provides a joint of 16 to 24 fibres by arraying them between two pin-positioning holes in the ferrule in a two-layer (two-row) arrangement. The connector includes a push-pull coupling mechanism and a ferrule spring loaded in the direction of the optical axis. The connector has a single male key which may be used to orient and limit the relative position between the connector and the component to which it is mated.

Connector interfaces are configured using a female plug without pins, a male plug with pins fixed and an adaptor as shown in Figure 1. The female plug is intermateable with the male plug. There are two angled-interface plugs, one called down-angled and the other up-angled. They are defined for both male and female plugs. The up and down descriptors refer to the tilt direction of the ferrule's angled end face relative to the fibre axis when looking toward the end face with the plug's key feature on the top. For down-angled plugs, the angled surface faces slightly downward. For up-angled plugs, the angled surface faces slightly upward. These different angles affect intermateability for the two adaptor types. An opposed keyway adaptor mates two plugs with the keys in opposite orientations, for example one side keyway-up and the other keyway-down. In contrast, an aligned keyway adaptor mates two plugs with the keys in the same orientation. When using an opposed keyway adaptor with angled interfaces, two down-angled plugs or two up-angled plugs shall be connected. For aligned keyway adaptors with angled interfaces, one down-angled plug and one up-angled plug shall be connected.

Moreover, connector interfaces between the female plug and the male plug are configured by applying a backplane housing and a printed board housing instead of the adaptor.

Additionally, the female plug interface is intermateable with the active device receptacle.

5 Interfaces

This document contains the following standard interfaces:

Interface IEC 61754-7-2-1: MPO female plug, down-angled interface for 16 to 24 fibres

Interface IEC 61754-7-2-2: MPO male plug, down-angled interface for 16 to 24 fibres

Interface IEC 61754-7-2-3: MPO adaptor interface – Opposed keyway configuration

Interface IEC 61754-7-2-4: MPO female plug, flat interface for 16 to 24 fibres

Interface IEC 61754-7-2-5: MPO male plug, flat interface for 16 to 24 fibres

Interface IEC 61754-7-2-6: MPO backplane housing interface

Interface IEC 61754-7-2-7: MPO printed board housing interface

Interface IEC 61754-7-2-8: MPO adaptor interface – Aligned keyway configuration

Interface IEC 61754-7-2-9: MPO active device receptacle, angled interface

Interface IEC 61754-7-2-10: MPO active device receptacle, flat interface

Interface IEC 61754-7-2-11: MPO female plug, up-angled interface for 16 to 24 fibres

Interface IEC 61754-7-2-12: MPO male plug, up-angled interface for 16 to 24 fibres

The interfaces showed in Table 1 are intermateable.

Table 1 – Intermateability between plugs and adapters/housings/receptacles

Female plugs	Adaptors/housings/receptacles	Male plugs
61754-7-2-1	61754-7-2-3	61754-7-2-2
61754-7-2-1	61754-7-2-8	61754-7-2-12
61754-7-2-11	61754-7-2-8	61754-7-2-2
61754-7-2-4	61754-7-2-3 and 61754-7-2-8	61754-7-2-5
61754-7-2-1 or 61754-7-2-11	61754-7-2-6 and 61754-7-2-7	61754-7-2-2 or 61754-7-2-12
61754-7-2-4	61754-7-2-6 and 61754-7-2-7	61754-7-2-5
61754-7-2-1	61754-7-2-9	N/A
61754-7-2-4	61754-7-2-10	N/A

NOTE Connector interfaces with 16 to 24 fibres will intermate and will correctly align the lower defined numbers of optical datum targets (see Figure 4).

Figure 1 shows MPO connector configurations.

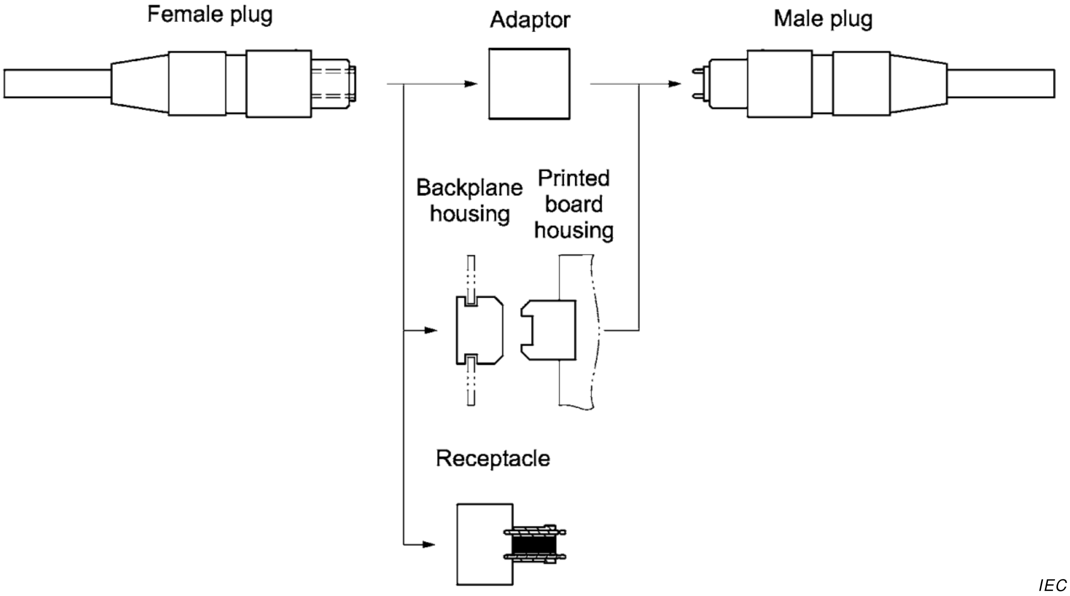


Figure 1 – MPO connector configurations

Figures 2 and 3 show the down-angled and up-angled interface of the MPO female plug. Table 2 gives the dimensions of the down- or up-angled interfaces of the MPO female plug.

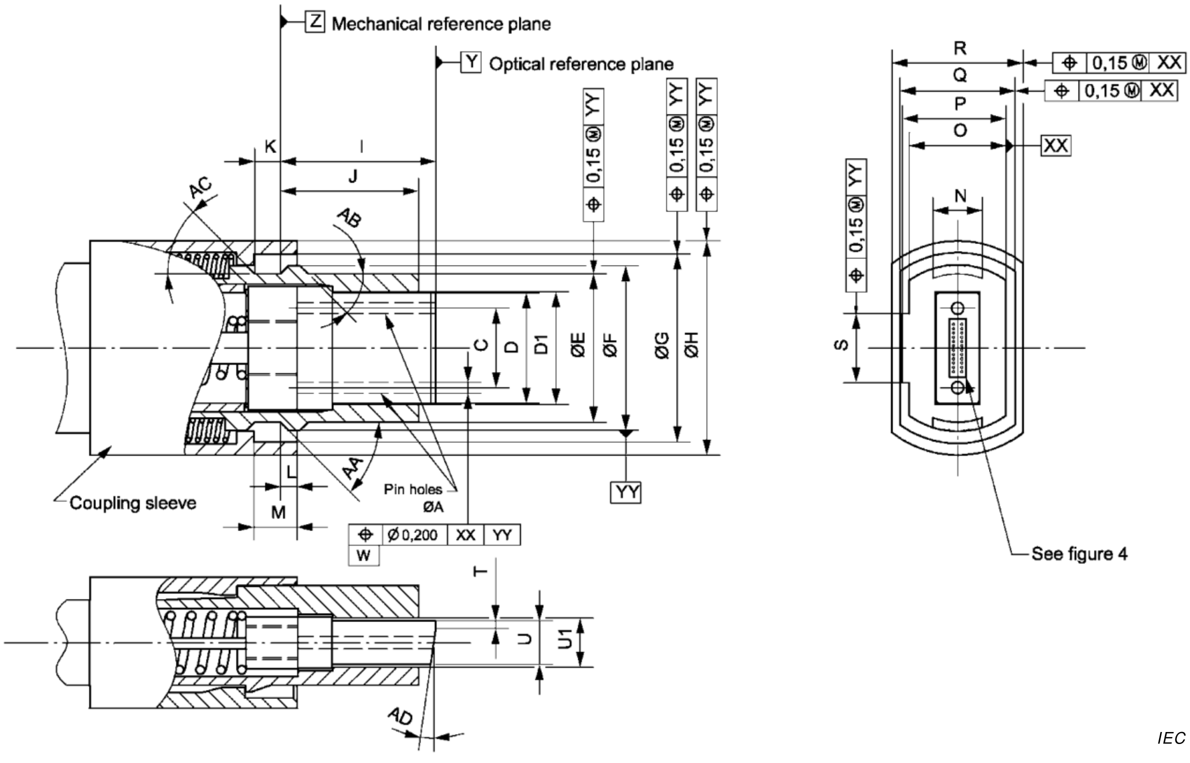


Figure 2 – MPO female plug, down-angled interface

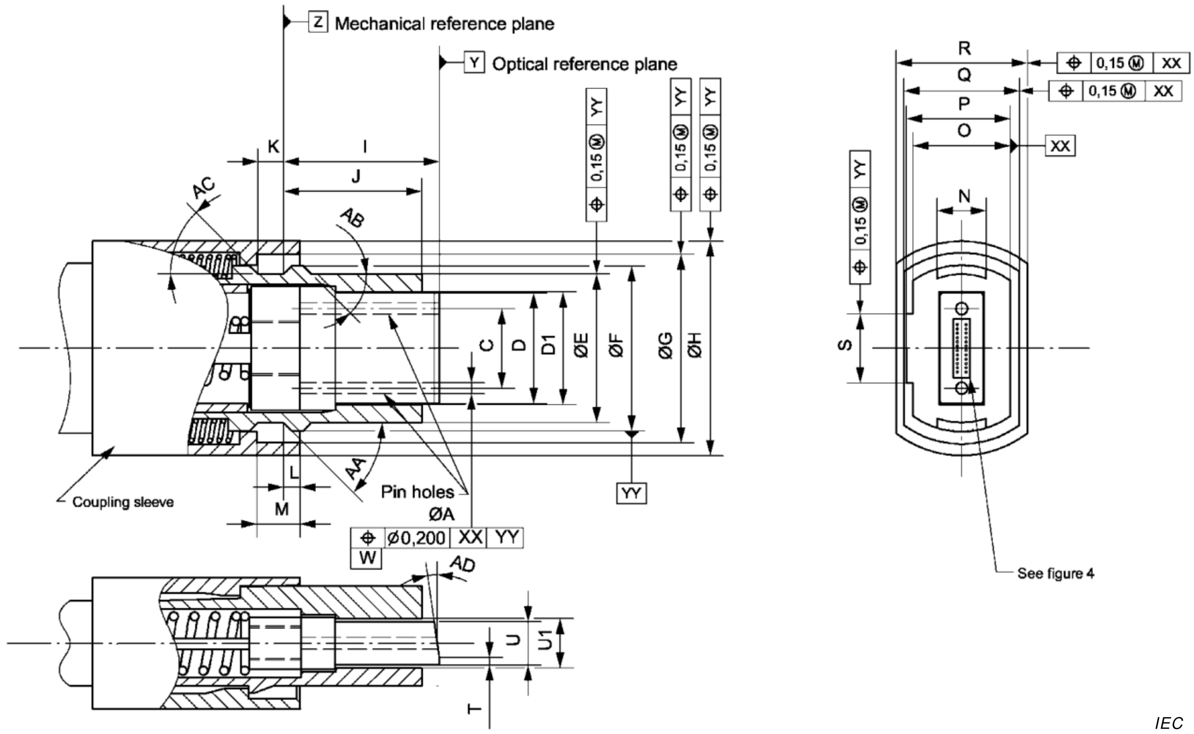


Figure 3 – MPO female plug, up-angled interface

Table 2 – Dimensions of the MPO female plug, down or up-angled interface

Reference	Dimensions		Remarks
	Minimum	Maximum	
A[a]	0,699 mm	0,701 mm	
C[b]	4,597 mm	4,603 mm	
D	6,3 mm	6,5 mm	
D1[g]	6,7 mm	-	
E	8,34 mm	8,54 mm	
F	9,49 mm	9,59 mm	
G	10,85 mm	11,05 mm	
H	12,19 mm	12,59 mm	
I[c,f]	8,8 mm	9,2 mm	
J	7,9 mm	8,1 mm	
K	1,4 mm	–	
L[d,e]	0,2 mm	0,8 mm	
M	2,4 mm	2,6 mm	
N	2,8 mm	3,0 mm	
O	4,89 mm	4,99 mm	
P	5,59 mm	5,69 mm	
Q	5,7 mm	–	
R	–	7,7 mm	
S	2,9 mm	3,1 mm	
T	–	0,8 mm	
U	2,4 mm	2,5 mm	
U1[g]	2,7 mm	-	
AA	42°	45°	
AB	–	45°	
AC	–	45°	
AD[h,i]	7,5°	8,5°	
The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.			

[a] Each pin-hole shall accept a gauge pin as shown in Figure 5 to a depth of 5,5 mm with a maximum force of 1,7 N. In addition, two pin-holes of a plug shall accept a gauge as shown in Figure 6 to a depth of 5,5 mm with a maximum force of 3,4 N.

[b] Dimension C is defined as the distance between two pin-hole centres.

[c] Dimension I is given for a fibre end face centre of a plug end when not mated. It is noticed that a ferrule is movable by a certain axial compression force, and therefore dimension I is variable. Ferrule compression force shall be 18,0 N to 22,0 N when a position of the fibre end face from datum Z is in the range of 8,2 mm to 8,4 mm.

[d] The coupling sleeve shall be movable by a certain axial compression force. Dimension L is given for a coupling sleeve end when not mated. Coupling sleeve compression force shall be 2,9 N to 6,9 N when the position of the coupling sleeve end face from datum Z is in the range of 0 mm to 0,1 mm to the right or to the left of datum Z.

[e] An adaptor coupling part shall be unlocked by a left-direction movement of a coupling sleeve, when it is separate from an adaptor. When the coupling sleeve is moved for unlocking, the position of the coupling sleeve end face shall be larger than 2,0 mm in the left direction from datum Z.

[f] Dimension I is defined at the centre line between the two pin-hole centres.

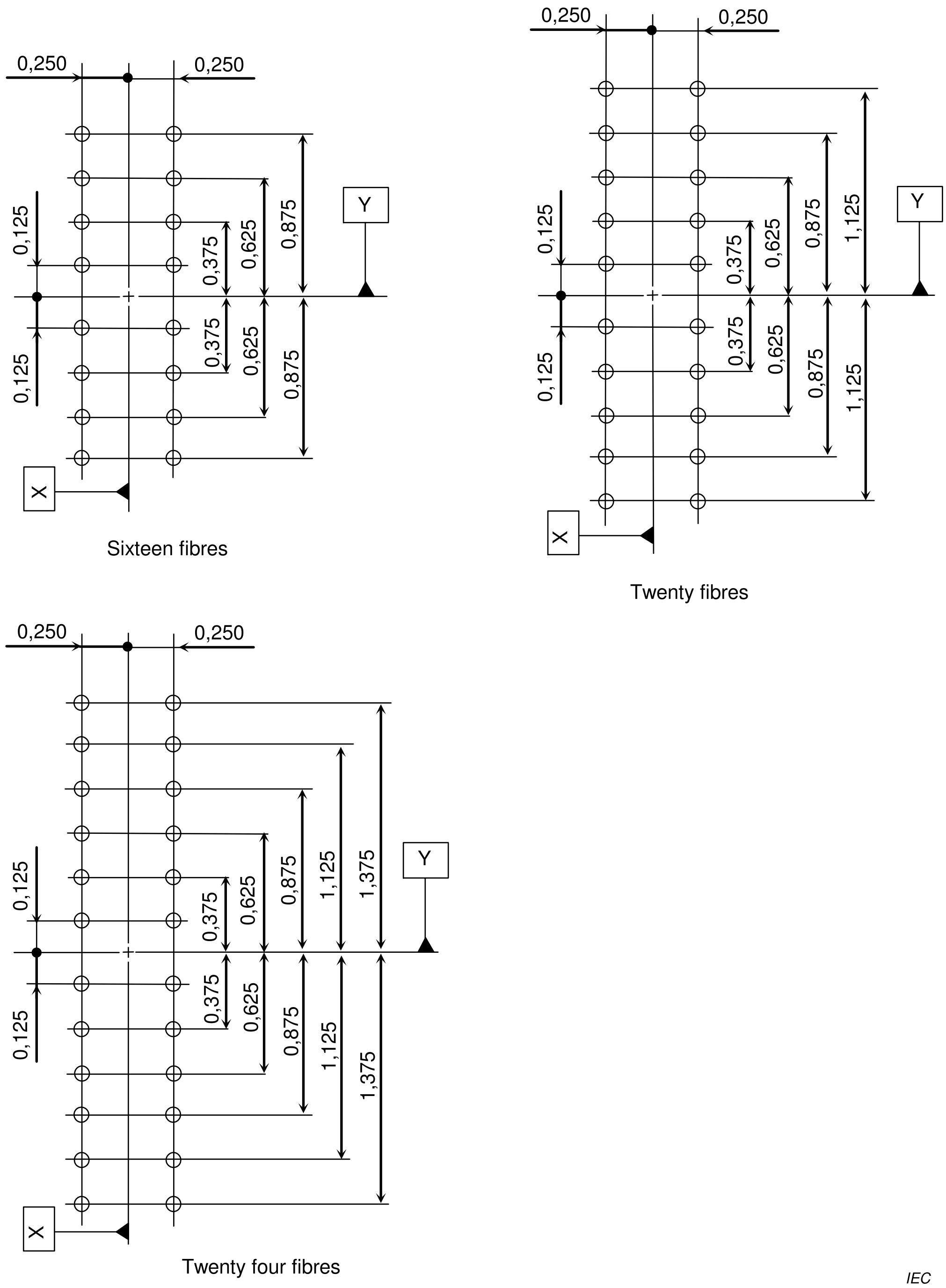
[g] Dimensions D1 and U1 are defined only at the end of the connector as shown.

[h] The down-angled and up-angled plugs shall be clearly marked to distinguish them from each other and flat interfaces through the use of colour, labelling, or other appropriate identification method. This identification method shall be visible when the plug is in the mated or unmated condition.

[i] Since angled MPO connectors require a Y-offset of the fibre holes in relation to the guide pin holes, and the Y-offset is referenced from the epoxy window of the ferrule, the angle shall be polished as a down-angle from the epoxy window. The orientation of the ferrule epoxy window may be reversed in the MPO connector to produce the up-angle variant.

Figure 4 shows the optical datum target location diagrams. Figure 5 shows the gauge pin, and its dimensions are given in Table 3.

Dimensions in millimetres



NOTE The optical datum target location diagram is shown in the figure. Here, datum X is defined as the line passing through two pin-hole centres, and datum Y is defined as the line perpendicular to datum X and passing through the midpoint of two pin-hole centres.

Figure 4 – Optical datum target location diagrams

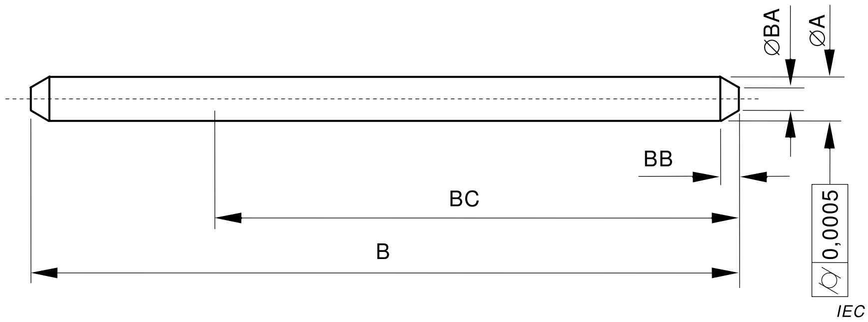


Figure 5 – Gauge pin

Table 3 – Dimensions of the gauge pin

Reference	Dimensions mm		Remarks
	Minimum	Maximum	
A[a]	0,698 5	0,699 0	
B[b]	10,8	11,2	
BA	0,2	0,4	
BB	0,2	0,5	
BC	6,0	–	

[a] Surface roughness R_z = 0,1 μm for the length of dimension BC.

[b] Typical dimensions.

Figure 6 shows the gauge for the plug, and its dimensions are given in Table 4.

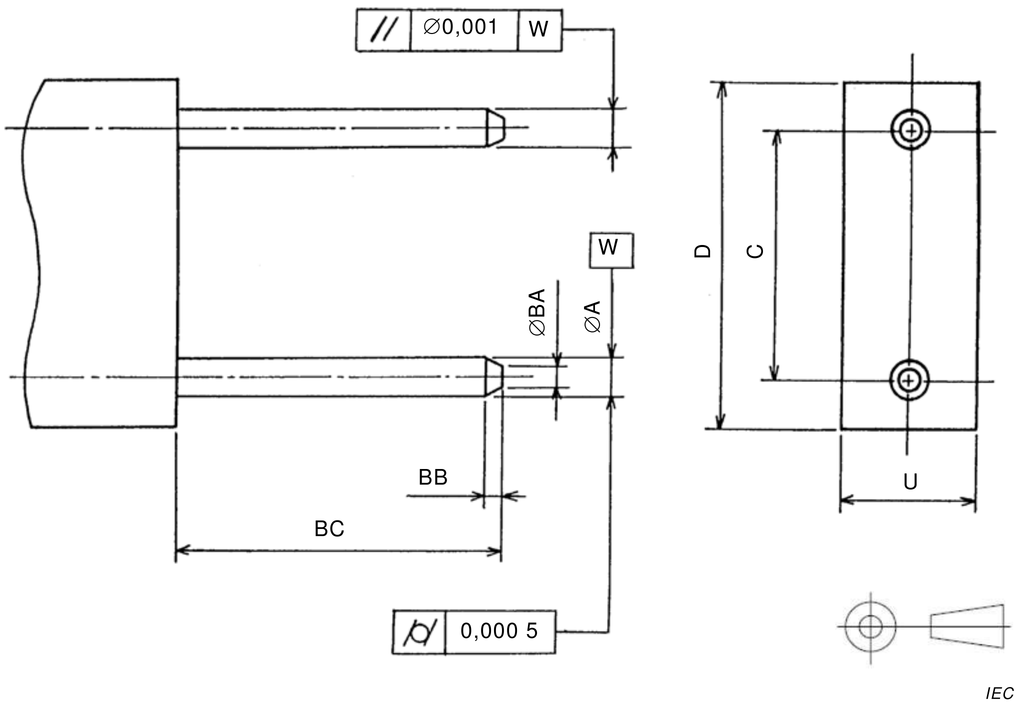


Figure 6 – Gauge for plug

Table 4 – Dimensions of the gauge for plug

Reference	Dimensions mm		Remarks
	Minimum	Maximum	
A[a]	0,698 5	0,699 0	
C	4,599 5	4,600 5	
D[b]	6,3	6,5	
U[b]	2,4	2,5	
BA	0,2	0,4	
BB	0,2	0,5	
BC	6,0	6,5	

[a] Surface roughness R_z = 0,1 µm.

[b] Typical dimensions.

Figure 7 and 8 show the down-angled and up-angled interface of the MPO male plug, respectively. Table 5 gives the dimensions of the down- or up-angled interfaces of the MPO male plug.

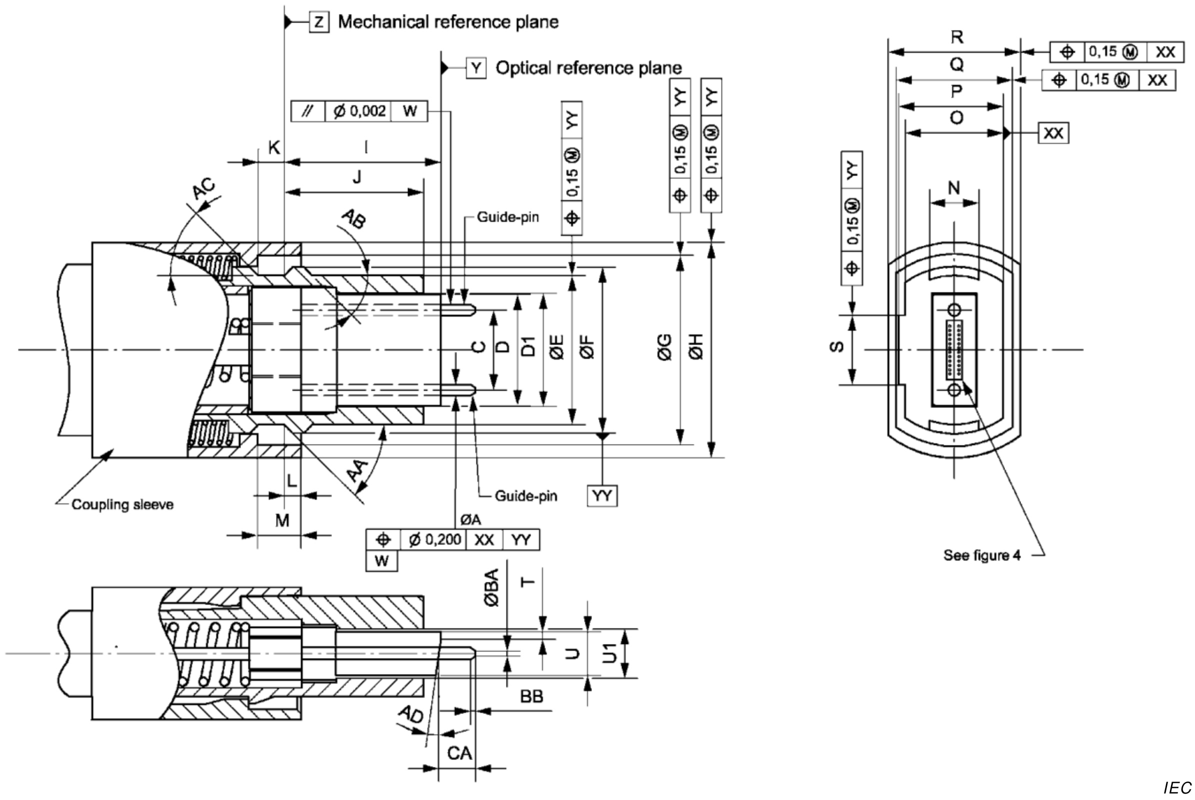


Figure 7 – MPO male plug, down-angled interface

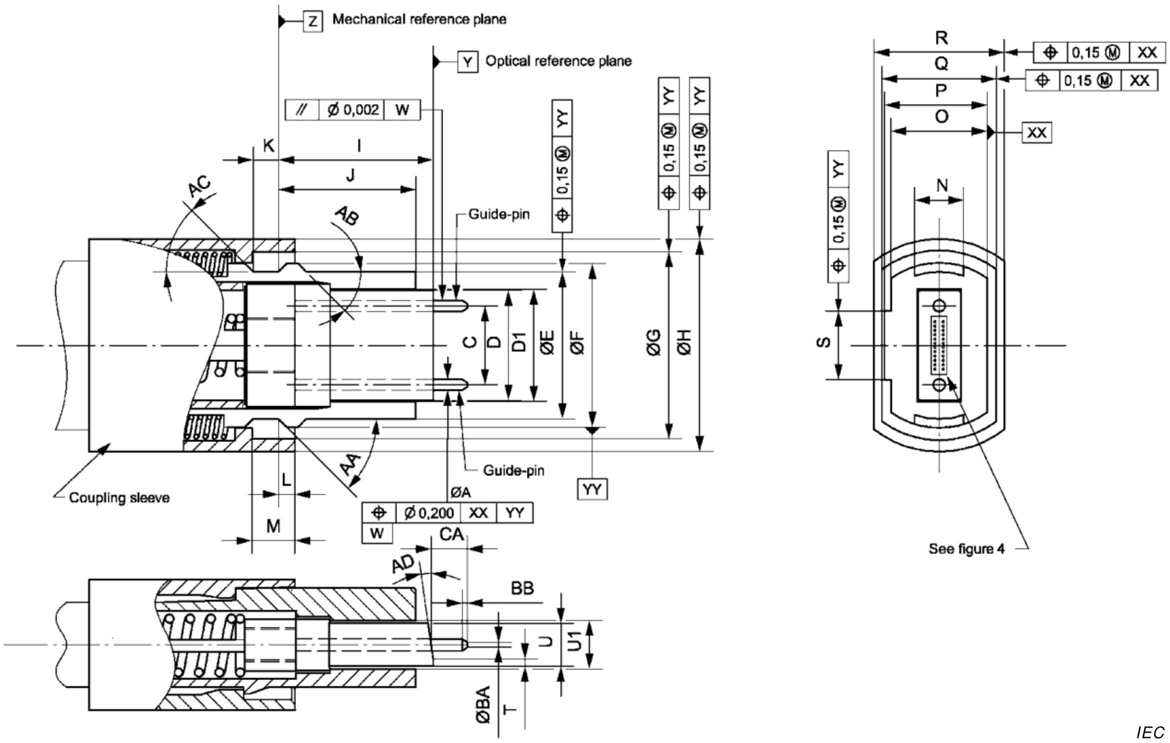


Figure 8 – MPO male plug, up-angled interface

Table 5 – Dimensions of the MPO male plug, down- or up-angled interface

Reference	Dimensions		Remarks
	Minimum	Maximum	
A[a]	0,697 mm	0,699 mm	
C[b]	4,597 mm	4,603 mm	
D	6,3 mm	6,5 mm	
D1[h]	6,7 mm	-	
E	8,34 mm	8,54 mm	
F	9,49 mm	9,59 mm	
G	10,85 mm	11,05 mm	
H	12,19 mm	12,59 mm	
I[c,g]	8,8 mm	9,2 mm	
J	7,9 mm	8,1 mm	
K	1,4 mm	–	
L[d,e]	0,2 mm	0,8 mm	
M	2,4 mm	2,6 mm	
N	2,8 mm	3,0 mm	
O	4,89 mm	4,99 mm	
P	5,59 mm	5,69 mm	
Q	5,7 mm	–	
R	–	7,7 mm	
S	2,9 mm	3,1 mm	
T	–	0,8 mm	
U	2,4 mm	2,5 mm	
U1[h]	2,7 mm	-	
AA	42º	45º	
AB	–	45º	
AC	–	45º	
AD[i,j]	7,5º	8,5º	
BA[f]	0,2 mm	0,4 mm	
BB	0,2 mm	0,5 mm	
CA	1,6 mm	3,3 mm	
The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.			

[a] Each guide pin shall be retained with a minimum force of 19,6 N. Surface roughness R_z shall be below 0,5 μm.

[b] Dimension C is defined as the distance between two guide-pin centres.

[c] Dimension I is given for a fibre end face centre of a plug end when not mated. It is noticed that a ferrule is movable by a certain axial compression force, and therefore the dimension I is variable. Ferrule compression force shall be 18,0 N to 22,0 N when a position of the fibre end face from datum Z is in the range of 8,2 mm to 8,4 mm.

[d] Coupling sleeve shall be movable by a certain axial compression force. Dimension L is given for a coupling sleeve end when not mated. Coupling sleeve compression force shall be 2,9 N to 6,9 N when a position of the coupling sleeve end face from datum Z is in the range of 0 mm to 0,1 mm to the right or to the left of datum Z.

[e] An adaptor coupling part shall be unlocked by a left-direction movement of a coupling sleeve, when it is separate from an adaptor. When the coupling sleeve is moved for unlocking, a position of the coupling sleeve end face shall be larger than 2,0 mm in the left direction from datum Z.

[f] The top shape of guide-pin may be a round shape that is symmetrical about the guide-pin axis with a minimum radius of 0,15 mm.

[g] Dimension I is defined at the centre line between the two guide-pin centres.

[h] Dimensions D1 and U1 are defined only at the end of the connector as shown.

[i] The down-angled and up-angled plugs shall be clearly marked to distinguish them from each other and flat interfaces through the use of colour, labelling, or other appropriate identification method. This identification method shall be visible when the plug is in the mated or unmated condition.

[j] Since angled MPO connectors require a Y-offset of the fibre holes in relation to the guide pin holes, and the Y-offset is referenced from the epoxy window of the ferrule, the angle shall be polished as a down-angle from the epoxy window. The orientation of the ferrule epoxy window may be reversed in the MPO connector to produce the up-angle variant.

Figure 9 shows the opposed keyway configuration of the MPO adaptor interface. Table 6 gives the dimensions of the MPO adaptor, opposed keyway configuration.

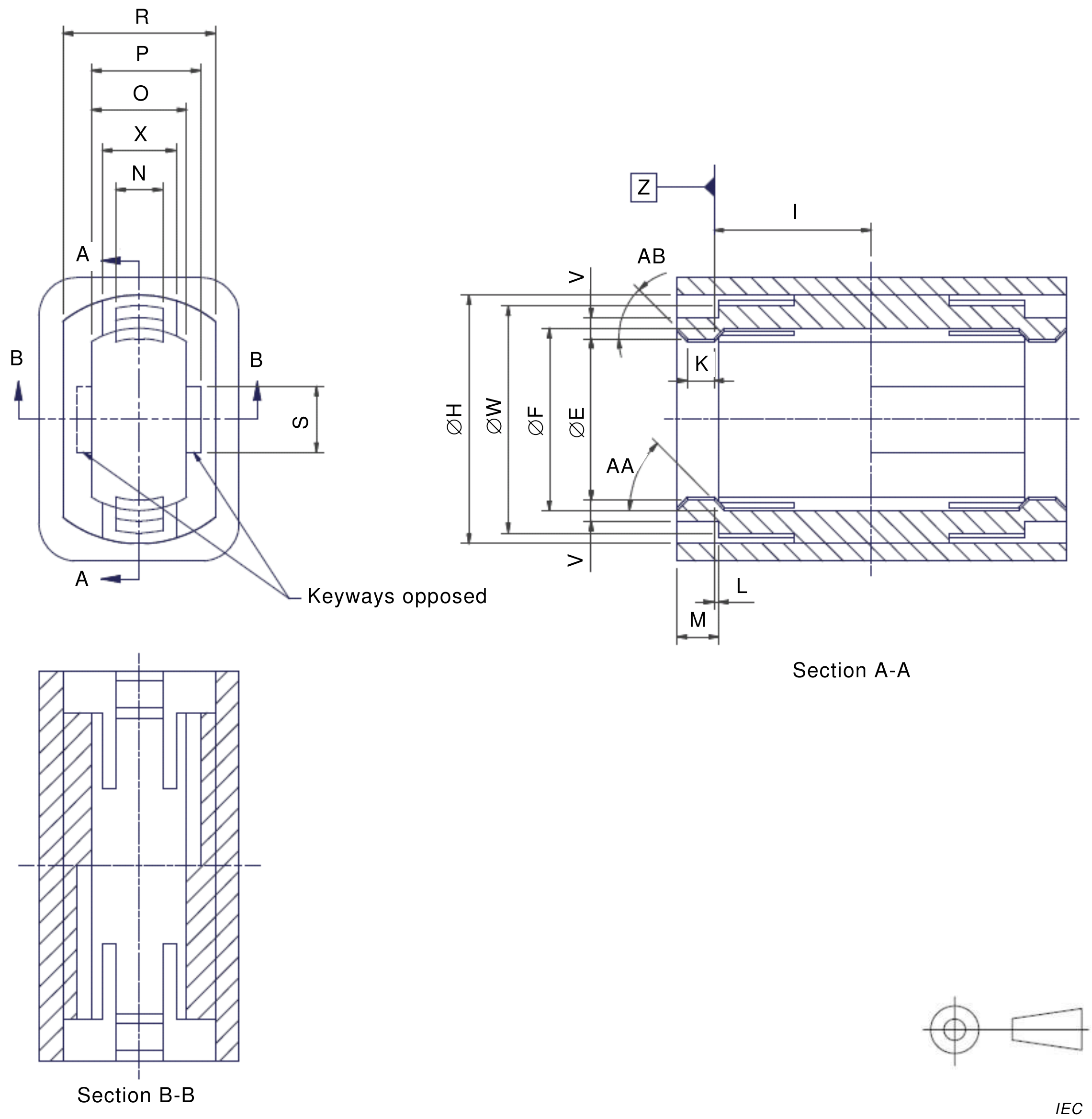


Figure 9 – MPO adaptor interface, opposed keyway configuration

Table 6 – Dimensions of the MPO adaptor interface, opposed keyway configuration

Reference	Dimensions mm		Remarks
	Minimum	Maximum	
E	8,54 mm	8,74 mm	
F	9,6 mm	9,7 mm	
H[a]	12,6 mm	–	
I	8,2 mm	8,4 mm	
K	–	1,39 mm	
L[b]	0	0,1 mm	
M	1,6 mm	2,0 mm	
N	2,4 mm	2,6 mm	
O	5,0 mm	5,1 mm	
P	5,7 mm	5,9 mm	
R	7,8 mm	–	
S	3,4 mm	3,6 mm	
V	0,95 mm	1,15 mm	
W[a]	-	12,2 mm	
X	3,4 mm	–	
AA	45°	48°	
AB	45°	50°	

The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.

[a] An adaptor latch shall be allowed the maximum deflection given by the connector and adaptor requirements. To ensure intermateability, the following requirement shall be met:

$$\frac{(\phi H - \phi W)}{2} > 0{,}53\ \text{mm}$$

[b] Dimension L is the distance from datum Z to a latch-ledge end. The latch-ledge end may be located to the right or to the left of datum Z.

Figure 10 shows the flat interface of the MPO female plug interface. Table 7 gives the dimensions of the flat interface of the MPO female plug.

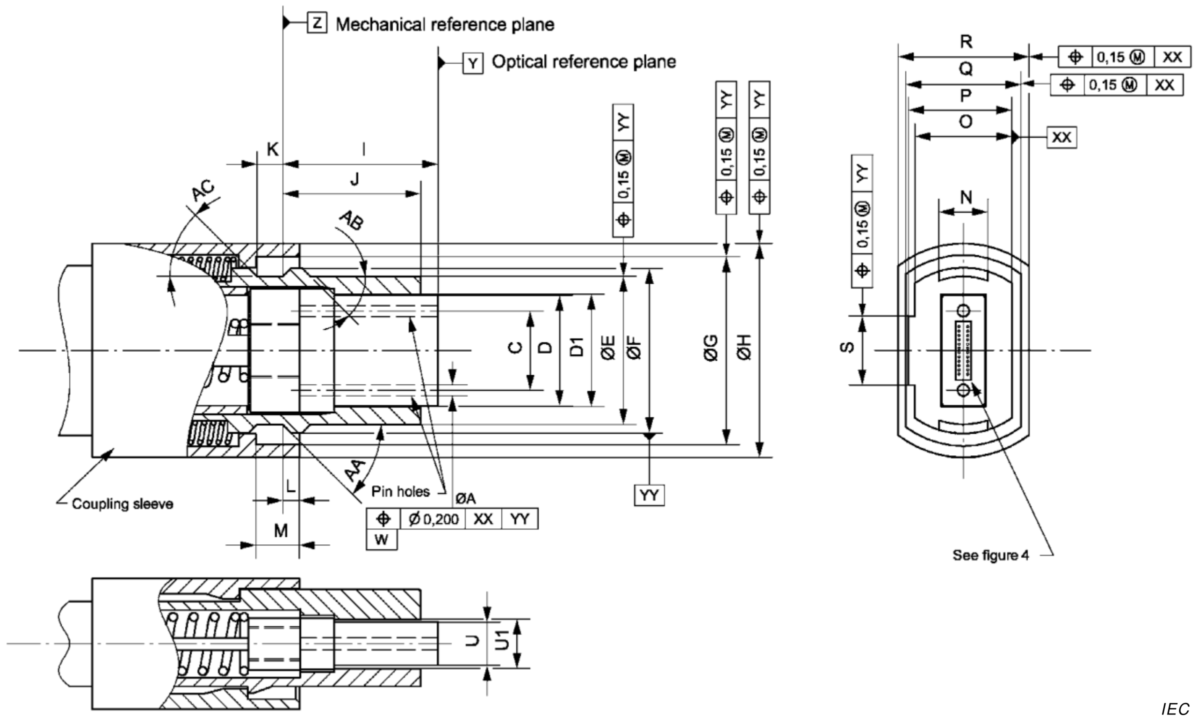


Figure 10 – MPO female plug, flat interface

Table 7 – Dimensions of the MPO female plug, flat interface

Reference	Dimensions		Remarks
	Minimum	Maximum	
A[a]	0,699 mm	0,701 mm	
C[b]	4,597 mm	4,603 mm	
D	6,3 mm	6,5 mm	
D1[g]	6,7 mm	-	
E	8,34 mm	8,54 mm	
F	9,49 mm	9,59 mm	
G	10,85 mm	11,05 mm	
H	12,19 mm	12,59 mm	
I[c,f]	8,8 mm	9,2 mm	
J	7,9 mm	8,1 mm	
K	1,4 mm	–	
L[d,e]	0,2 mm	0,8 mm	
M	2,4 mm	2,6 mm	
N	2,8 mm	3,0 mm	
O	4,89 mm	4,99 mm	
P	5,59 mm	5,69 mm	
Q	5,7 mm	–	
R	–	7,7 mm	
S	2,9 mm	3,1 mm	
U	2,4 mm	2,5 mm	
U1[g]	2,7 mm	-	
AA	42°	45°	
AB	–	45°	
AC	–	45°	

The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.

[a] Each pin-hole shall accept a gauge pin as shown in Figure 5 to a depth of 5,5 mm with a maximum force of 1,7 N. In addition, two pin-holes of a plug shall accept a gauge as shown in Figure 6 to a depth of 5,5 mm with a maximum force of 3,4 N.

[b] Dimension C is defined as the distance between two pin-hole centres.

[c] Dimension I is given for a fibre end face centre of a plug end when not mated. It is noticed that a ferrule is movable by a certain axial compression force, and therefore the dimension I is variable. Ferrule compression force shall be 18,0 N to 22,0 N when a position of the fibre end face from datum Z is in the range of 8,2 mm to 8,4 mm.

[d] The coupling sleeve shall be movable by a certain axial compression force. Dimension L is given for a coupling sleeve end when not mated. Coupling sleeve compression force shall be 2,9 N to 6,9 N when the position of the coupling sleeve end face from datum Z is in the range of 0 mm to 0,1 mm to the right or to the left of datum Z.

[e] An adaptor coupling part shall be unlocked by a movement away from the adaptor of a coupling sleeve, when it is separate from the adaptor. When the coupling sleeve is moved for unlocking, the position of the coupling sleeve end face shall be larger than 2,0 mm in the left direction from datum Z.

[f] Dimension I is defined at the centre line between the two pin-hole centres.

[g] Dimensions D1 and U1 are defined only at the end of the connector as shown.

Figure 11 shows the flat interface of the MPO male plug. Table 8 gives the dimensions of the flat interface of the MPO male plug.

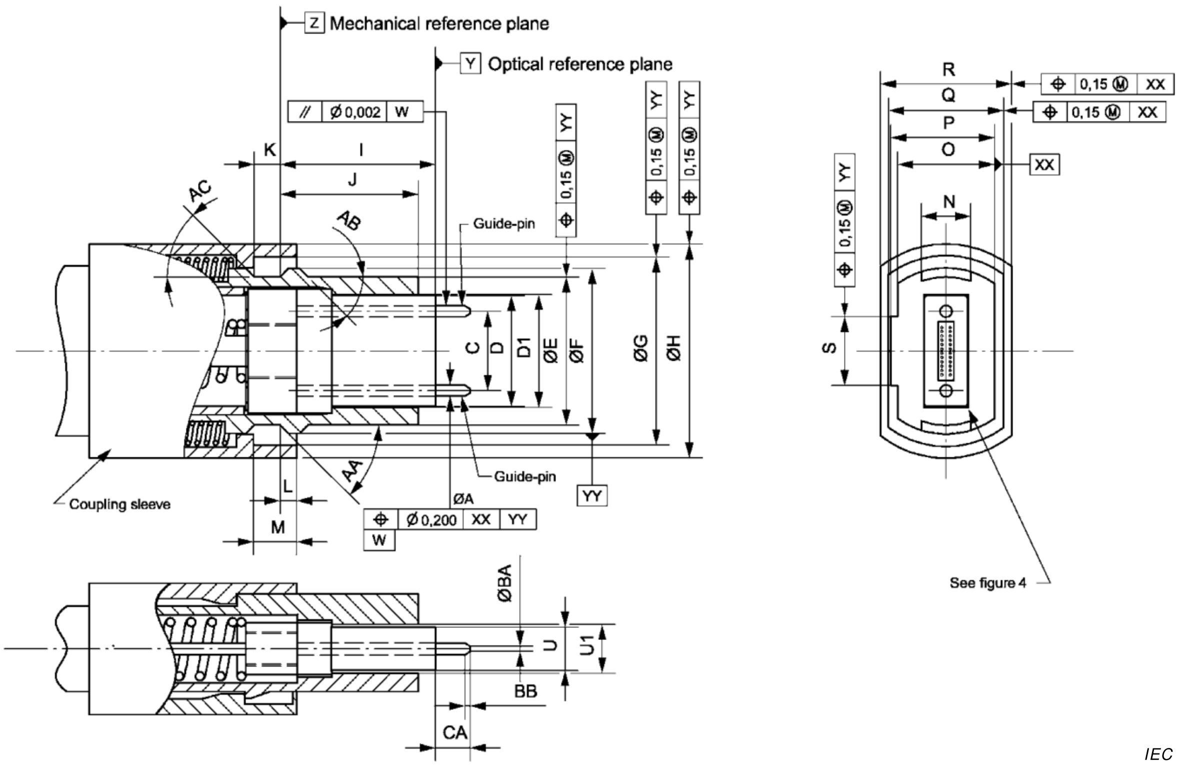


Figure 11 – MPO male plug, flat interface

Table 8 – Dimensions of the MPO male plug, flat interface

Reference	Dimensions		Remarks
	Minimum	Maximum	
A[a]	0,697 mm	0,699 mm	
C[b]	4,597 mm	4,603 mm	
D	6,3 mm	6,5 mm	
D1[h]	6,7 mm	-	
E	8,34 mm	8,54 mm	
F	9,49 mm	9,59 mm	
G	10,85 mm	11,05 mm	
H	12,19 mm	12,59 mm	
I[c,g]	8,8 mm	9,2 mm	
J	7,9 mm	8,1 mm	
K	1,4 mm	–	
L[d,e]	0,2 mm	0,8 mm	
M	2,4 mm	2,6 mm	
N	2,8 mm	3,0 mm	
O	4,89 mm	4,99 mm	
P	5,59 mm	5,69 mm	
Q	5,7 mm	–	
R	–	7,7 mm	
S	2,9 mm	3,1 mm	
U	2,4 mm	2,5 mm	
U1[h]	2,7 mm	-	
AA	42°	45°	
AB	–	45°	
AC	–	45°	
BA[f]	0,2 mm	0,4 mm	
BB	0,2 mm	0,5 mm	
CA	1,6 mm	3,3 mm	

The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.

[a] Each guide-pin shall be retained with a minimum force of 19,6 N. Surface roughness R_z shall be below 0,5 μm.

[b] Dimension C is defined as the distance between two guide-pin centres.

[c] Dimension I is given for a fibre end face centre of a plug end when not mated. It is noticed that a ferrule is movable by a certain axial compression force, and therefore the dimension I is variable. Ferrule compression force shall be 18,0 N to 22,0 N when a position of the fibre end face from datum Z is in the range of 8,2 mm to 8,4 mm.

[d] The coupling sleeve shall be movable by a certain axial compression force. Dimension L is given for a coupling sleeve end when not mated. Coupling sleeve compression force shall be 2,9 N to 6,9 N when the position of the coupling sleeve end face from datum Z is in the range of 0 mm to 0,1 mm to the right or to the left of datum Z.

[e] An adaptor coupling part shall be unlocked by a movement away from the adaptor of a coupling sleeve, when it is separate from the adaptor. When the coupling sleeve is moved for unlocking, the position of the coupling sleeve end face shall be larger than 2,0 mm in the left direction from datum Z.

[f] The top shape of guide-pin may be a round shape that is symmetrical about the guide-pin axis with a minimum radius of 0,15 mm.

[g] Dimension I is defined at the centre line between the two guide-pin centres.

[h] Dimensions D1 and U1 are defined only at the end of the connector as shown.

Figure 12 shows the MPO backplane housing interface. Table 9 gives the dimensions of the MPO backplane housing interface.

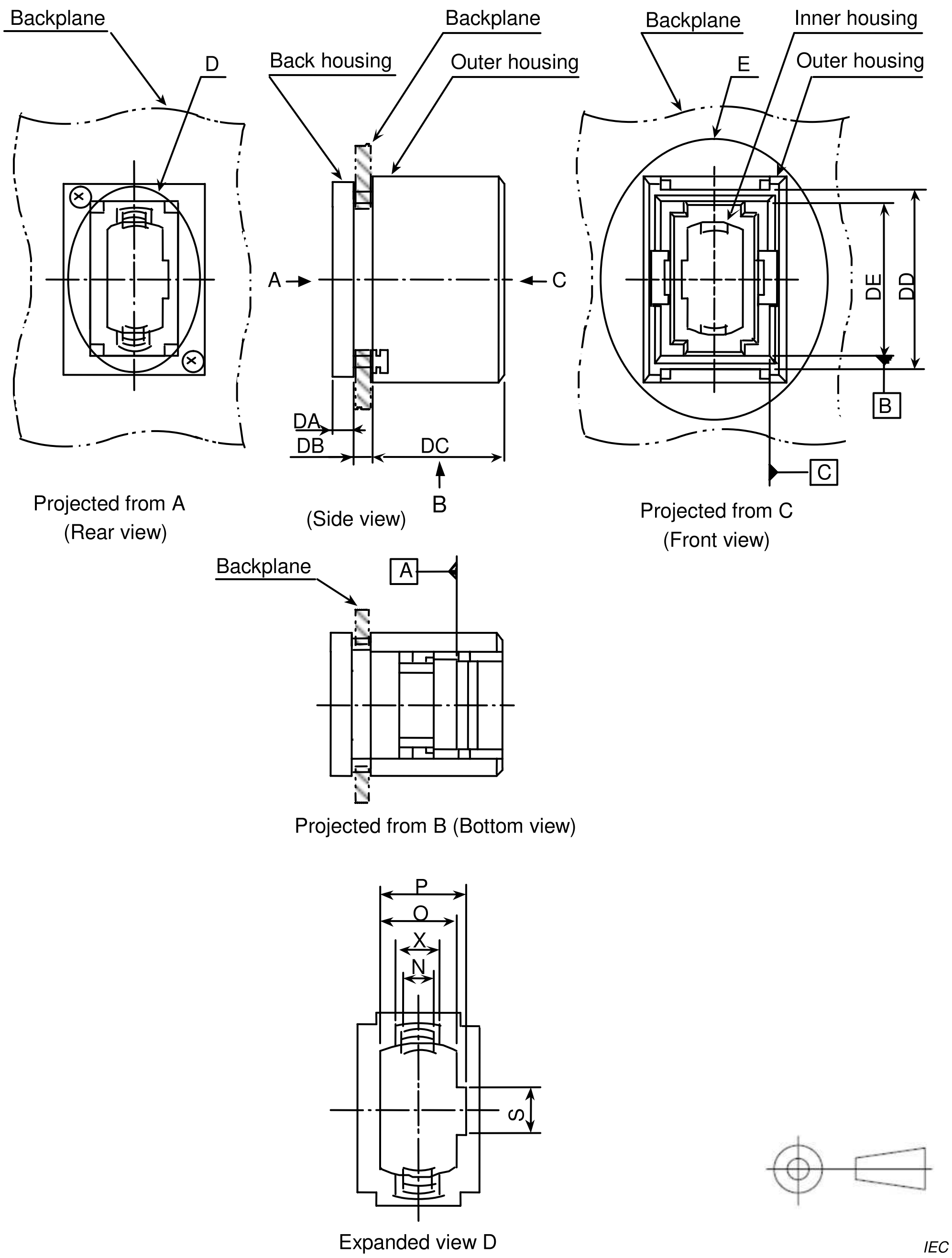
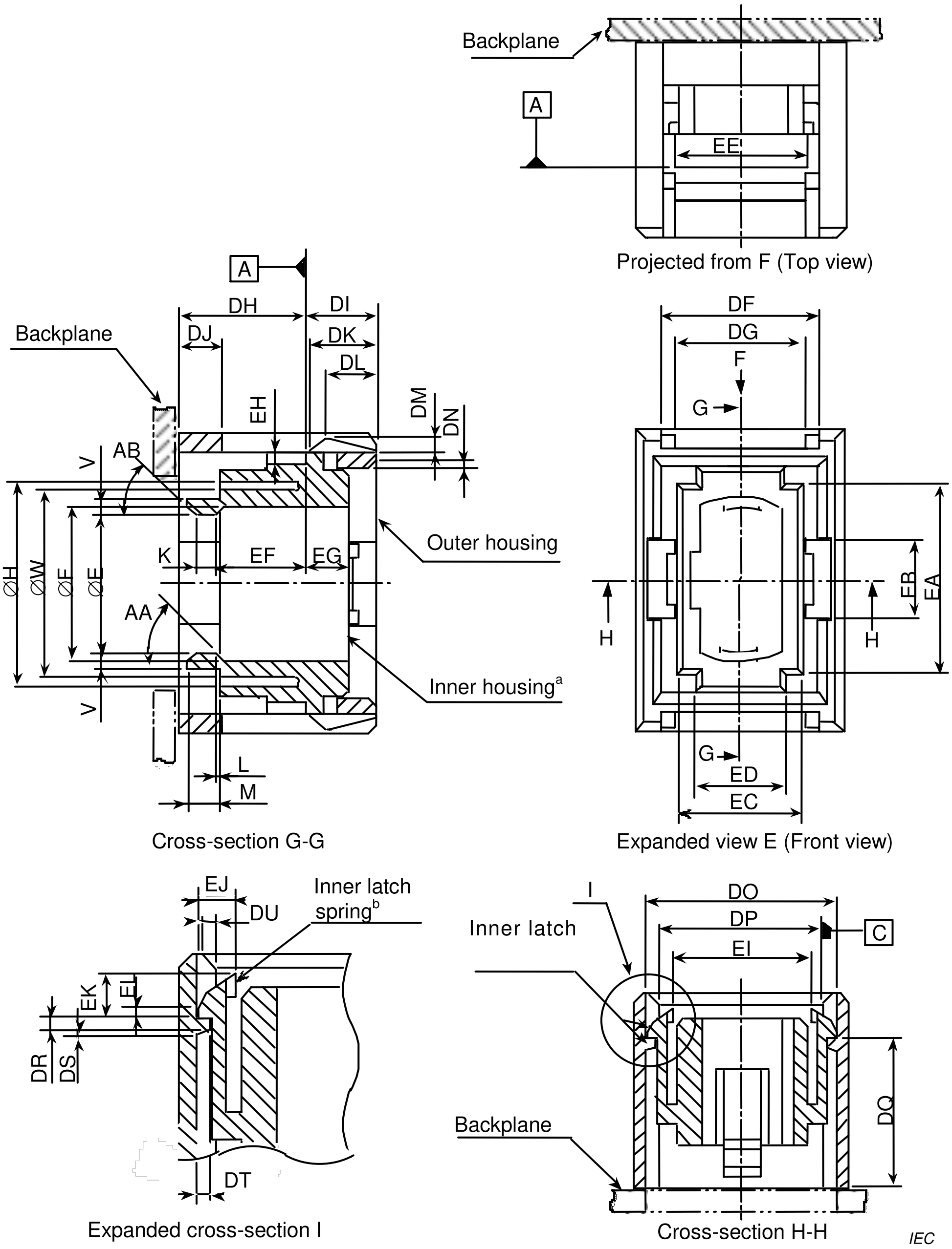


Figure 12 – MPO backplane housing interface *(1 of 2)*



[a] In the figure of cross-section G-G, the inner housing shall be movable to the right at least 0,9 mm under the condition that the inner latch is completed. In addition, the inner housing shall be movable at least 2 mm to the left when the inner latch is released.

[b] In the figure of expanded cross-section I, the inner latch spring shall be moved by more than 0,65 mm to the right when the inner latch is released or latched.

Figure 12 *(2 of 2)*

Table 9 – Dimensions of the MPO backplane housing

Reference	Dimensions		Remarks
	Minimum	Maximum	
E	8,54 mm	8,74 mm	A part of diameter
F	9,6 mm	9,7 mm	A part of diameter
H	12,6 mm	–	A part of diameter
K	1,19 mm	1,39 mm	
L	0	0,1 mm	
M	1,6 mm	2,0 mm	
N	2,4 mm	2,6 mm	
O	5,0 mm	5,1 mm	
P	5,7 mm	5,9 mm	
S	3,4 mm	3,6 mm	
V	0,95 mm	1,15 mm	
W	11,8 mm	12,2 mm	A part of diameter
X	3,4 mm	–	
AA	45°	48°	
AB	45°	50°	
DA			See Table 10
DB			See Table 10
DC	12,25 mm	12,35 mm	
DD	16,5 mm	16,6 mm	
DE	14,3 mm	14,4 mm	
DF	9,91 mm	10,01 mm	
DG	8,2 mm	8,4 mm	
DH[a]	7,9 mm	8,1 mm	
DI[a]	4,15 mm	4,45 mm	
DJ	2,65 mm	2,75 mm	
DK	4,1 mm	4,3 mm	
DL	3,35 mm	3,45 mm	
DM	0,9 mm	1,0 mm	
DN	0,55 mm	0,65 mm	
DO	11,55 mm	11,65 mm	
DP	9,91 mm	10,01 mm	
DQ	9,15 mm	9,25 mm	
DR	0,35 mm	0,45 mm	
DS	0,25 mm	0,35 mm	
DT	0,55 mm	0,65 mm	
DU	0,55 mm	0,70 mm	
EA	12,14 mm	12,2 mm	
EB	4,95 mm	5,05 mm	
EC	7,94 mm	8,00 mm	
ED	5,6 mm	5,8 mm	
EE	8,15 mm	8,25 mm	

Reference	Dimensions		Remarks
	Minimum	Maximum	
EF	5,55 mm	5,65 mm	
EG	2,55 mm	2,65 mm	
EH	0,85 mm	0,95 mm	
EI	8,6 mm	8,7 mm	
EJ	1,45 mm	1,55 mm	
EK	1,9 mm	2,0 mm	
EL	0,35 mm	0,45 mm	

[a] These dimensions are given when the inner housing is moved in its most left-side position on condition that the inner latch is engaged.

Table 10 gives the grade.

Table 10 – Grade

Grade	Reference	Dimensions mm		Remarks
		Minimum	Maximum	
1[a]	DA DB	2,0 2,65	2,1 2,75	Backplane thickness 2,4 mm
2[a]	DA DB	2,0 3,45	2,1 3,55	Backplane thickness 3,2 mm

[a] Add grade number to the interface reference number.

Figure 13 shows the MPO printed board housing interface. Table 11 gives the dimensions of the MPO printed board housing interface.

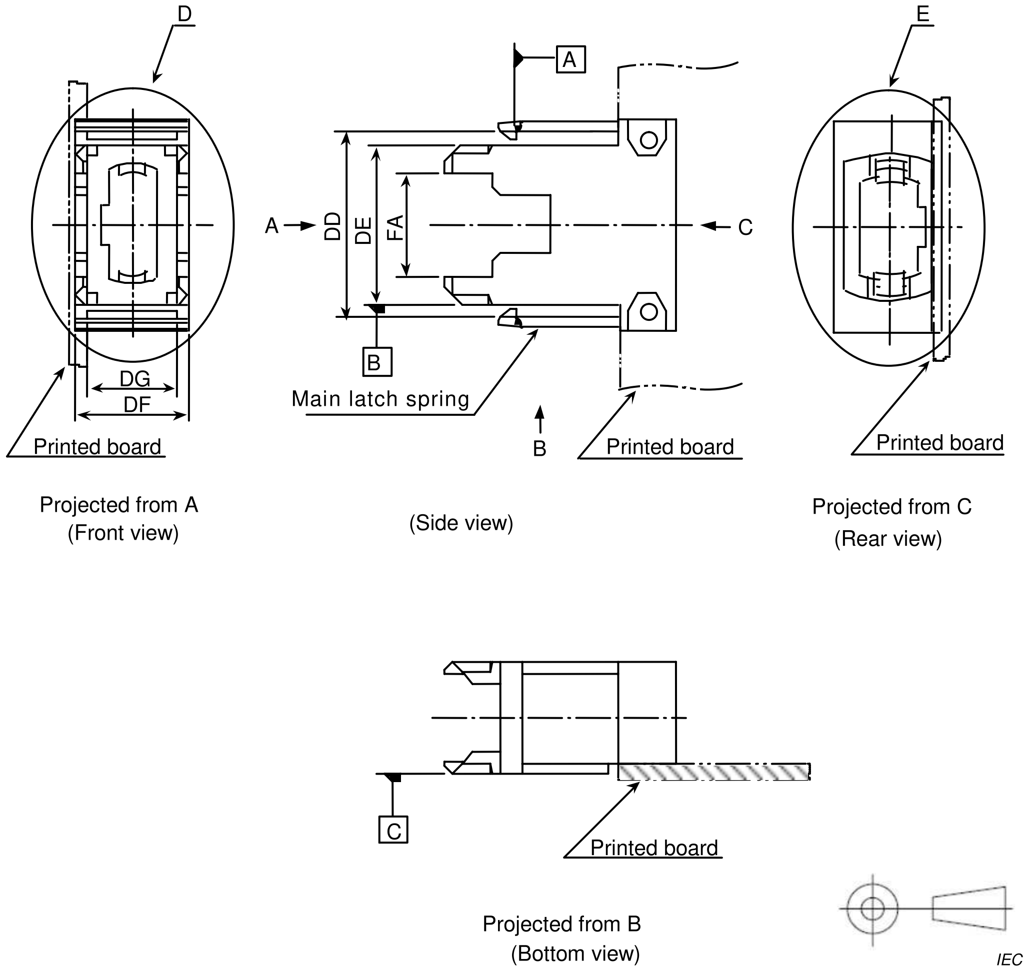


Figure 13 – MPO printed board housing interface *(1 of 2)*

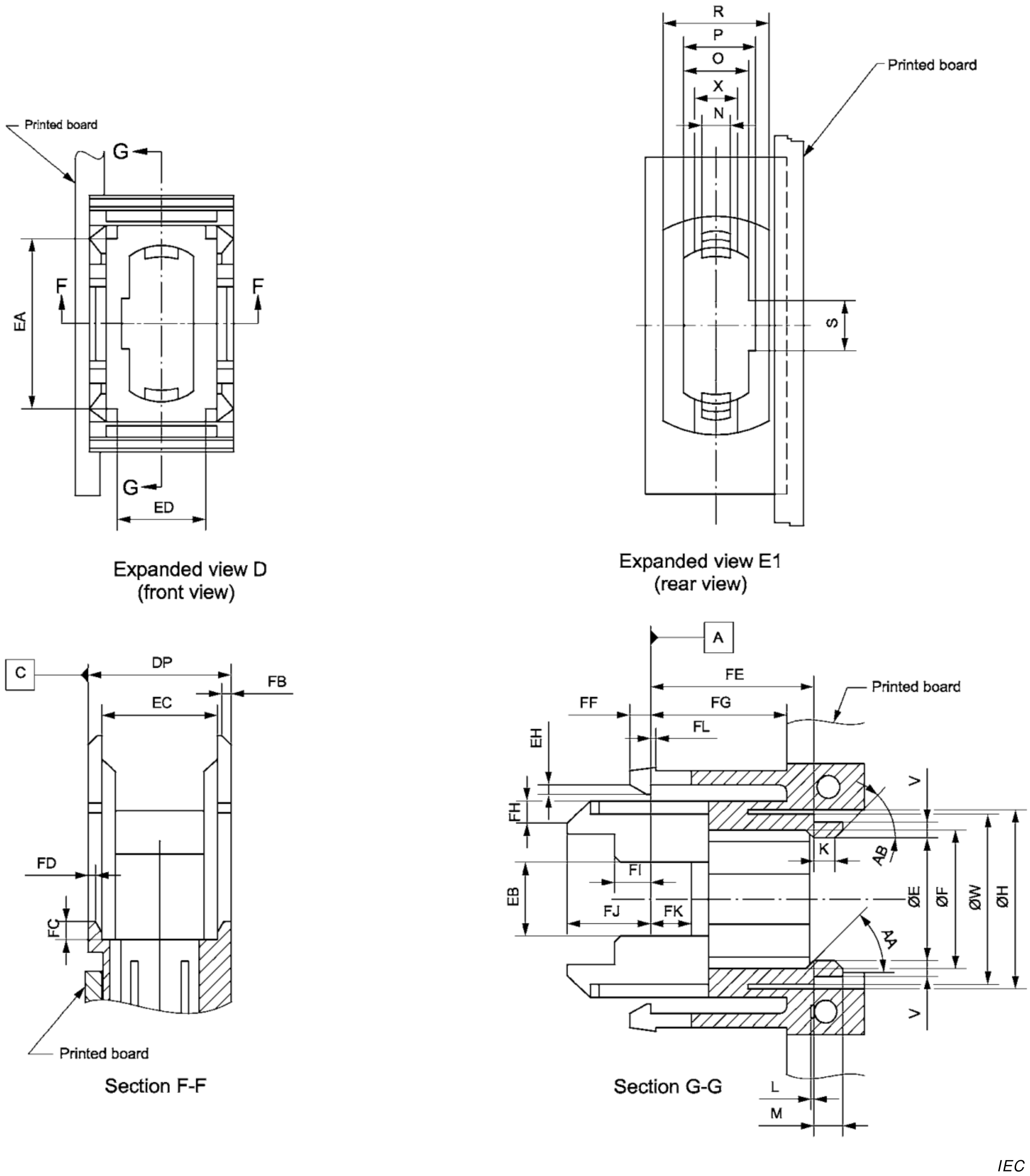


Figure 13 *(2 of 2)*

Table 11 – Dimensions of the MPO printed board housing interface

Reference	Dimensions		Remarks
	Minimum	Maximum	
E	8,54 mm	8,74 mm	A part of diameter
F	9,6 mm	9,7 mm	A part of diameter
H	12,6 mm	–	A part of diameter
K	1,19 mm	1,39 mm	
L	0	0,1 mm	
M	1,6 mm	2,0 mm	
N	2,4 mm	2,6 mm	
O	5,0 mm	5,1 mm	
P	5,7 mm	5,9 mm	
R	7,8 mm	–	
S	3,4 mm	3,6 mm	
V	0,95 mm	1,15 mm	
W	11,8 mm	12,2 mm	A part of diameter
X	3,4 mm	–	
AA	45°	48°	
AB	45°	50°	
DD[a]	16,5 mm	16,6 mm	
DE	14, 05 mm	14,15 mm	
DF	9,8 mm	9,9 mm	
DG	7,9 mm	8,1 mm	
DP	9,8 mm	9,9 mm	
EA	12,21 mm	12,27 mm	
EB	5,1 mm	5,2 mm	
EC	8,01 mm	8,07 mm	
ED	5,9 mm	6,1 mm	
EH	0,75 mm	0,85 mm	
FA	9,1 mm	9,3 mm	
FB	0,6 mm	0,8 mm	
FC	1,35 mm	1,65 mm	
FD	0,25 mm	0,35 mm	
FE	11,05 mm	11,15 mm	
FF	1,55 mm	1,65 mm	
FG	8,9 mm	9,0 mm	
FH	1,4 mm	1,6 mm	
FI	1,9 mm	2,2 mm	
FJ	6,35 mm	6,55 mm	
FK	2,9 mm	3,0 mm	
FL	0,35 mm	0,45 mm	

[a] The dimension DD is defined at the top of the main latch spring. The dimension DD shall become greater than 18,6 mm when the printed board housing is coupled to, or removed from, a backplane housing.

Figure 14 shows the aligned keyway configuration of the MPO adaptor interface. Table 12 gives dimensions of the MPO adaptor interface, aligned keyway configuration.

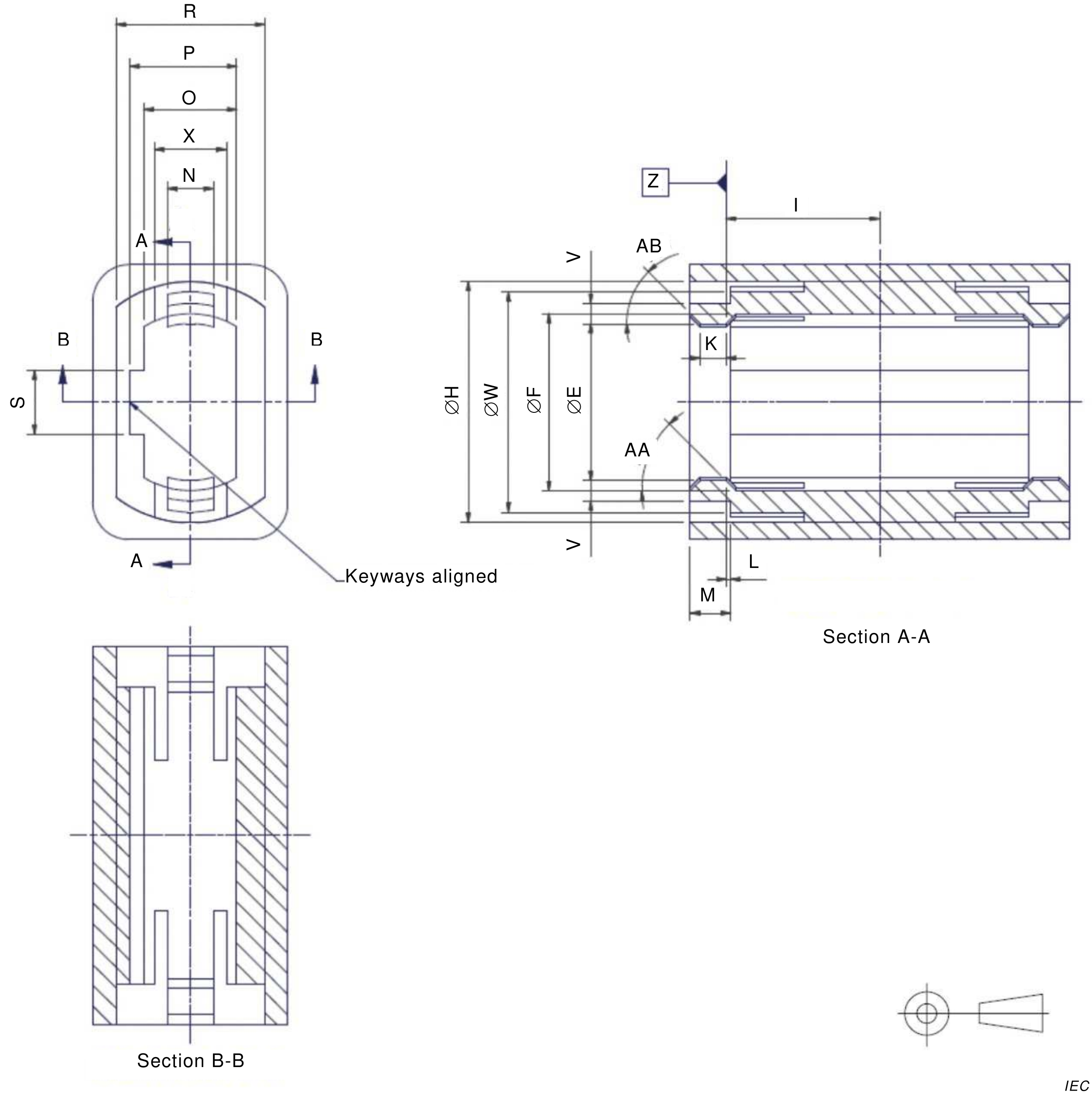


Figure 14 – MPO adaptor interface, aligned keyway configuration

Table 12 – Dimensions of the MPO adaptor interface, aligned keyway configuration

Reference	Dimensions		Remarks
	Minimum	Maximum	
E	8,54 mm	8,74 mm	
F	9,6 mm	9,7 mm	
H[a]	12,6 mm	–	
I[c]	8,2 mm	8,4 mm	
K	–	1,39 mm	
L[b]	0	0,1 mm	
M	1,6 mm	2,0 mm	
N	2,4 mm	2,6 mm	
O	5,0 mm	5,1 mm	
P	5,7 mm	5,9 mm	
R	7,8 mm	–	
S	3,4 mm	3,6 mm	
V	0,95 mm	1,15 mm	
W[a]	11,8 mm	12,2 mm	
X	3,4 mm	–	
AA	45°	48°	
AB	45°	50°	

The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.

[a] An adaptor latch shall be allowed the maximum deflection given by the connector and adaptor requirements. To ensure intermateability, the following requirement shall be met:

$$\frac{(\phi H - \phi W)}{2} > 0{,}53\ \text{mm}$$

[b] Dimension L is the distance from datum Z to a latch-ledge end. The latch-ledge end may be located to the right or to the left of datum Z.

[c] The keyways for each MPO plug are aligned on the same side of the adaptor. Dimension I is the distance from datum Z to the adaptor centreline.

Figure 15 shows the angled interface of the MPO active device receptacle, and Table 13 gives dimensions of the angled interface of the MPO active device receptacle.

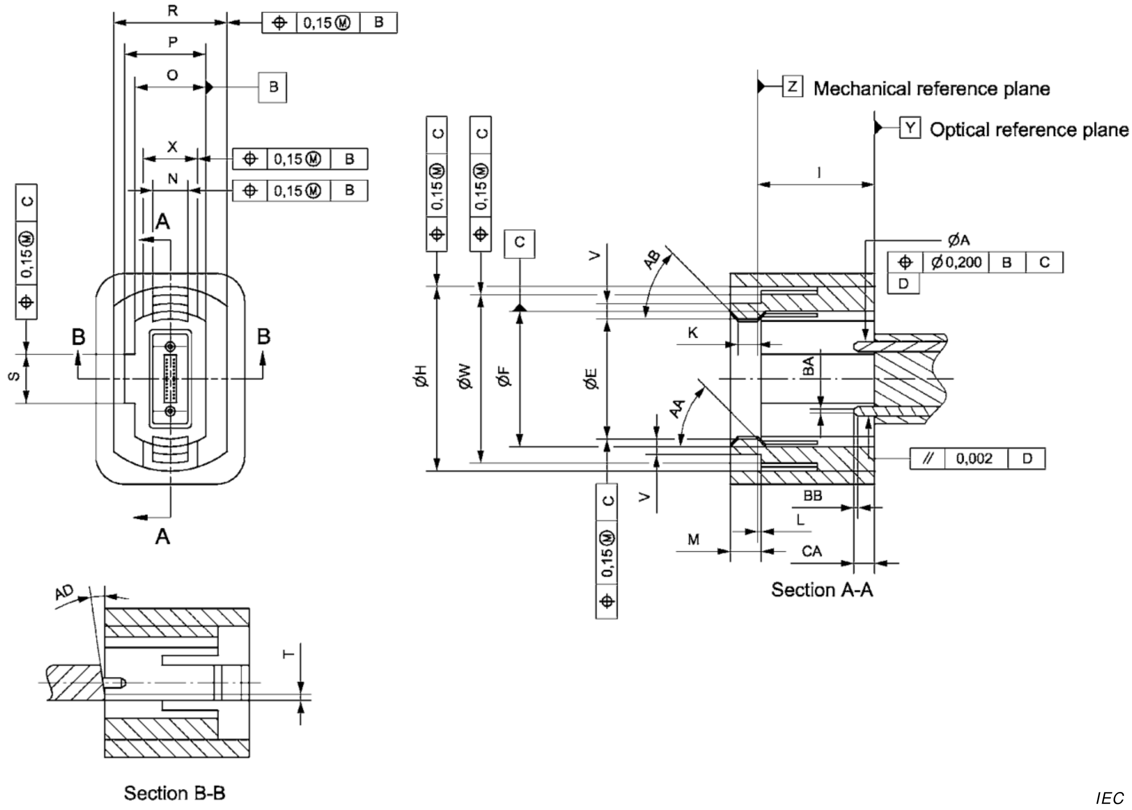


Figure 15 – MPO active device receptacle, angled interface

Table 13 – Dimensions of the MPO active device receptacle, angled interface

Reference	Dimensions mm		Remarks
	Minimum	Maximum	
E	8,54 mm	8,74 mm	
F	9,6 mm	9,7 mm	
H[a]	12,6 mm	–	
I[c]	8,2 mm	8,4 mm	
K	–	1,39 mm	
L[b]	0	0,1 mm	
M	1,6 mm	2,0 mm	
N	2,4 mm	2,6 mm	
O	5,0 mm	5,1 mm	
P	5,7 mm	5,9 mm	
R	7,8 mm	–	
S	3,4 mm	3,6 mm	
T	-	0,8 mm	
V	0,95 mm	1,15 mm	
W[a]	-	12,2 mm	
X	3,4 mm	–	
AA	45°	48°	
AB	45°	50°	
AD	7,5°	8,5°	
BA	0,2 mm	0,4 mm	
BB	0,2 mm	0,5 mm	
CA	1,6 mm	3,3 mm	

The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.

[a] An adaptor latch shall be allowed the maximum deflection given by the connector and adaptor requirements. To ensure intermateability, the following requirement shall be met:

$$\frac{(\phi H - \phi W)}{2} > 0{,}53\ \text{mm}$$

[b] Dimension L is the distance from datum Z to a latch-ledge end. The latch-ledge end may be located to the right or to the left of datum Z.

[c] Dimension I is defined at the centre line between the two guide-pin centres.

Figure 16 shows the flat interface of the MPO active device receptacle, and Table 14 gives dimensions of the flat interface of the MPO active device receptacle.

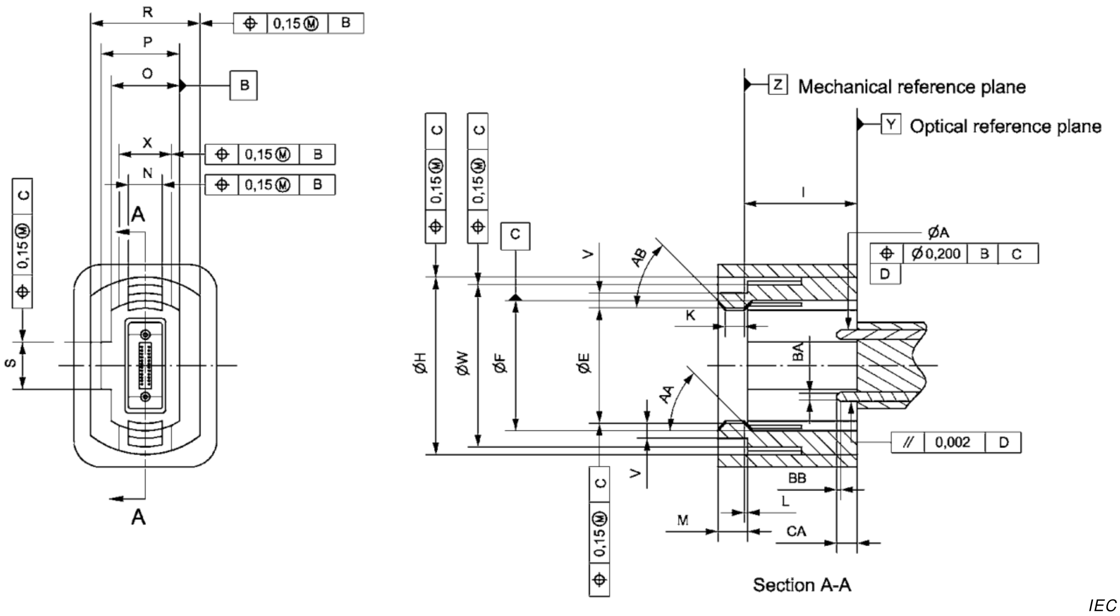


Figure 16 – MPO active device receptacle, flat interface

Table 14 – Dimensions of the MPO active device receptacle, flat interface

Reference	Dimensions mm		Remarks
	Minimum	Maximum	
E	8,54 mm	8,74 mm	
F	9,6 mm	9,7 mm	
H[a]	12,6 mm	–	
I[c]	8,2 mm	8,4 mm	
K	–	1,39 mm	
L[b]	0	0,1 mm	
M	1,6 mm	2,0 mm	
N	2,4 mm	2,6 mm	
O	5,0 mm	5,1 mm	
P	5,7 mm	5,9 mm	
R	7,8 mm	–	
S	3,4 mm	3,6 mm	
V	0,95 mm	1,15 mm	
W[a]	-	12,2 mm	
X	3,4 mm	–	
Y	1,6 mm	3,3 mm	
AA	45°	48°	
AB	45°	50°	
BA	0,2 mm	0,4 mm	
BB	0,2 mm	0,5 mm	
CA	1,6 mm	3,3 mm	

The mating/unmating force between an MPO connector and adaptor shall not exceed 40,0 N.

[a] An adaptor latch shall be allowed the maximum deflection given by the connector and adaptor requirements. To ensure intermateability, the following requirement shall be met:

$$\frac{(\phi H - \phi W)}{2} > 0{,}53 \text{ mm}$$

[b] Dimension L is the distance from datum Z to a latch-ledge end. The latch-ledge end may be located to the right or to the left of datum Z.

[c] Dimension I is defined at the centre line between the two guide-pin centres.

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